

[Soal] Geometry Everyday Project

AZZAM LABIB HAKIM |INSTAGRAM: @HAXUV_WORLD |THREADS: @AZZAM_29_12

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§1 Soal

§1.1 Korea MO 2022, Problem 1

Let ABC be an acute triangle with circumcenter O , and let D , E , and F be the feet of altitudes from A , B , and C to sides BC , CA , and AB , respectively. Denote by P the intersection of the tangents to the circumcircle of ABC at B and C . The line through P perpendicular to EF meets AD at Q , and let R be the foot of the perpendicular from A to EF . Prove that DR and OQ are parallel.

§1.2 Japan MO 2022, Problem 3

In an isosceles triangle ABC , $AB = AC$. Take a point O inside the triangle (not on the boundary). Draw a circle ω with center O and passing through C . Suppose $\omega \cap BC = \{C, D\}$, $\omega \cap AC = \{C, E\}$. Denote the circumcircle of $\triangle AEO$ as Γ . And suppose $\Gamma \cap \omega = \{E, F\}$. Prove that the circumcenter of $\triangle BDF$ is on Γ .

§1.3 IMO 2014, Problem 4

Let P and Q be on segment BC of an acute triangle ABC such that $\angle PAB = \angle BCA$ and $\angle CAQ = \angle ABC$. Let M and N be the points on AP and AQ , respectively, such that P is the midpoint of AM and Q is the midpoint of AN . Prove that the intersection of BM and CN is on the circumference of triangle ABC .

§1.4 USA January TST for IMO 2016, Problem 3

Let ABC be an acute scalene triangle and let P be a point in its interior. Let A_1 , B_1 , C_1 be projections of P onto triangle sides BC , CA , AB , respectively. Find the locus of points P such that AA_1 , BB_1 , CC_1 are concurrent and $\angle PAB + \angle PBC + \angle PCA = 90^\circ$.

§1.5 Singapore MO Open 2022

For $\triangle ABC$ and its circumcircle ω , draw the tangents at B, C to ω meeting at D . Let the line AD meet the circle with center D and radius DB at E inside $\triangle ABC$. Let F be the point on the extension of EB and G be the point on the segment EC such that $\angle AFB = \angle AGE = \angle A$. Prove that the tangent at A to the circumcircle of $\triangle AFG$ is parallel to BC .

§1.6 Iran-Ukraine Camp 2021 Exam I P-1

Incircle of the triangle $\triangle ABC$ is tangent to sides BC, AC, AB at points D, E, F respectively. Let I be the incentre of triangle and K be the intersection point of AI, BC . Circumcircle w of the triangle $\triangle IDK$ intersects IB and IC at points P and Q . Prove that lines EF, PQ and the tangent to w from D are concurrent.

§1.7 USAMO 2015, Problem 2

Quadrilateral $APBQ$ is inscribed in circle ω with $\angle P = \angle Q = 90^\circ$ and $AP = AQ < BP$. Let X be a variable point on segment $\overline{PQ}$. Line AX meets ω again at S (other than A). Point T lies on arc AQB of ω such that $\overline{XT}$ is perpendicular to $\overline{AX}$. Let M denote the midpoint of chord $\overline{ST}$. As X varies on segment $\overline{PQ}$, show that M moves along a circle.

§1.8 All-Russian 2022 Grade 11 Problem 3

An acute-angled triangle ABC is fixed on a plane with largest side BC . Let PQ be an arbitrary diameter of its circumscribed circle, and the point P lies on the smaller arc AB , and the point Q is on the smaller arc AC . Points X, Y, Z are feet of perpendiculars dropped from point P to the line AB , from point Q to the line AC and from point A to line PQ . Prove that the center of the circumscribed circle of triangle XYZ lies on a fixed circle.

Terjemahan: Diberikan sebuah segitiga lancip ABC yang tetap di sebuah bidang dengan sisi terpanjangnya adalah BC . Misalkan PQ adalah sebuah diameter sembarang dari lingkaran luar segitiga ABC , dengan titik P berada di busur AB yang lebih kecil dan titik Q berada di busur AC yang lebih kecil. Titik X, Y, Z berturut-turut adalah kaki tinggi titik P ke garis AB , titik Q ke garis AC , dan titik A ke garis PQ . Buktikan bahwa pusat lingkaran luar segitiga XYZ berada pada sebuah lingkaran tetap.

§1.9 All-Russian 2022 Grade 9 Problem 2

Given triangle ABC with incenter I and A -excenter J . Circle ω_b centered at point O_b passes through point B and is tangent to line CI at point I . Circle ω_c with center O_c passes through point C and touches line BI at point I . Let O_bO_c and IJ intersect at point K . Find the ratio IK/KJ .

§1.10 CGMO 2022, Problem 3

In triangle ABC , $AB > AC$, I is the incenter, AM is the median. The line through I and is perpendicular to BC intersect AM at point L , and the symmetry of I with respect to point A is J . Prove that: $\angle ABJ = \angle LBI$.

§1.11 IMSO 2021

In the diagram below, $AB = BC = CD$, $\angle ABC = 135^\circ$ and $\angle BCD = 105^\circ$. What is the measure, in degrees, of $\angle ADC$?

§1.12 Philippine MO 2022, Problem 6

In $\triangle ABC$, let D be the point on side BC such that $AB + BD = DC + CA$. The line AD intersects the circumcircle of $\triangle ABC$ again at point $X \neq A$. Prove that one of the common tangents of the circumcircles of $\triangle BD X$ and $\triangle CD X$ is parallel to BC .

§1.13 Finnish National High School Competition 2019, Problem 3

Let $ABCD$ be a cyclic quadrilateral whose side AB is at the same time the diameter of the circle. The lines AC and BD intersect at point E and the extensions of lines AD and BC intersect at point F . Segment EF intersects the circle at G and the extension of segment EF intersects AB at H . Show that if G is the midpoint of FH , then E is the midpoint of GH .

§1.14 Finnish National High School Competition 2001, Problem 1

In the right triangle ABC , CF is the altitude based on the hypotenuse AB . The circle centered at B and passing through F and the circle with centre A and the same radius intersect at a point of CB . Determine the ratio $FB : BC$.

§1.15 Iberoamerican 2022, Problem 5

Let ABC be an acute triangle with circumcircle Γ . Let P and Q be points in the half plane defined by BC containing A , such that BP and CQ are tangents to Γ and $PB = BC = CQ$. Let K and L be points on the external bisector of the angle $\angle CAB$, such that $BK = BA$, $CL = CA$. Let M be the intersection point of the lines PK and QL . Prove that $MK = ML$.

§1.16 Random Problem by Wildan Bagus Wicaksono

Diberikan persegi panjang $ABCD$. Titik P adalah perpotongan garis BC dengan garis yang melalui A dan tegak lurus AC . Titik Q pada segmen CD . Garis PQ memotong garis AD di R . Garis BR memotong garis AQ di X . Buktikan bahwa jika titik Q bergerak sepanjang segmen CD , maka besar $\angle BXC$ konstan.

(Authored by Wildan Bagus Wicaksono)

§1.17 USAJMO 2023, Problem 2

In an acute triangle ABC , let M be the midpoint of $\overline{BC}$. Let P be the foot of the perpendicular from C to AM . Suppose that the circumcircle of triangle ABP intersects line BC at two distinct points B and Q . Let N be the midpoint of $\overline{AQ}$. Prove that $NB = NC$.

§1.18 Korea 2023, Problem 1

In a triangle ABC ($\overline{AB} < \overline{AC}$), points $D(\neq A, B)$ and $E(\neq A, C)$ lies on side AB and AC respectively. Point P satisfies $\overline{PB} = \overline{PD}$, $\overline{PC} = \overline{PE}$. $X(\neq A, C)$ is on the arc AC of the circumcircle of triangle ABC not including B . Let $Y(\neq A)$ be the intersection of circumcircle of triangle ADE and line XA . Prove that $\overline{PX} = \overline{PY}$.

§1.19 IMO 2023, Problem 2

Let ABC be an acute-angled triangle with $AB < AC$. Let Ω be the circumcircle of ABC . Let S be the midpoint of the arc CB of Ω containing A . The perpendicular from A to BC meets BS at D and meets Ω again at $E \neq A$. The line through D parallel to BC meets line BE at L . Denote the circumcircle of triangle BDL by ω . Let ω meet Ω again at $P \neq B$. Prove that the line tangent to ω at P meets line BS on the internal angle bisector of $\angle BAC$.

§1.20 Japan 2023, Problem 2

In an acute triangle ABC , let D, E, F be the midpoints of BC, CA, AB respectively. Let X, Y be the feet of altitude from D to AB, AC respectively. The line through F parallel to XY meets line DY at P . Prove that $AD \perp EP$.

§1.21 Random Problem 2

Diberikan segitiga ABC dengan besar sudut $\angle ABC = 60^\circ$ dan $\angle BCA = 100^\circ$. Pada sisi AB dan AC berturut-turut terdapat titik D dan E sehingga $\angle EDC = 2\angle BCD = 2\angle CAB$. Tentukan besar sudut $\angle BED$.

§1.22 LMNAS SMP 33 Penyisihan nomor 16

Diberikan $\triangle ABC$ dengan $AB = AC$. Lingkaran dalam $\triangle ABC$ menyinggung sisi AB dan BC berturut-turut di P dan Q . Garis CP memotong lingkaran dalam $\triangle ABC$ di titik P dan R . Jika $PQ = 4$ dan $PR = 5$, panjang QR adalah ...

- a. $\frac{1}{2}\sqrt{14}$ b. 2 c. $\frac{3}{5}\sqrt{14}$ d. $\frac{4}{5}\sqrt{14}$ e. 3

§1.23 LMNAS SMP 33 Penyisihan nomor 3 isian singkat

Diberikan segitiga tumpul ABC dengan $AB = BC$. Titik D berada di dalam segitiga tersebut sedemikian sehingga $AD = BD$, $\angle ADB = 140^\circ$, dan $\angle ADC = 150^\circ$. Besar sudut ACD dalam satuan $^\circ$ (derajat) adalah . . .

§1.24 LMNAS SMP 31 Semifinal nomor 17

Diberikan segitiga lancip ABC . Garis $\ell_1, \ell_2 \perp AB$ berturut-turut pada A dan B . Misalkan M titik tengah AB . Garis melalui M tegak lurus AC dan BC memotong ℓ_1 dan ℓ_2 berturut-turut pada E dan F . Misalkan D titik potong dari EF dan MC . Jika $EF = \frac{64}{3}$, $FM = 16$, dan $BD = 12$. Luas segitiga AMF dapat dinyatakan dalam $a\sqrt{b}$ dimana a dan b bilangan asli dan b tidak dapat dibagi oleh bilangan kuadrat sempurna selain 1. Nilai dari $a + b$ adalah ...

§1.25 Pelatda Pra OSP SMA 2024 DKI Jakarta

Pada segitiga ABC , titik D adalah kaki garis tinggi dari C . Titik E dan F pada AC dan BC , berturut-turut $AE = AD$ dan $BF = BD$. Titik S adalah refleksi C pada titik pusat lingkaran luar $\triangle ABC$. Buktikan bahwa $SE = SF$.

§1.26 IMO 2024, Problem 4

Let ABC be a triangle with $AB < AC < BC$. Let the incenter and incircle of triangle ABC be I and ω , respectively. Let X be the point on line BC different from C such that the line through X parallel

to AC is tangent to ω . Similarly, let Y be the point on line BC different from B such that the line through Y parallel to AB is tangent to ω . Let AI intersect the circumcircle of triangle ABC at $P \neq A$. Let K and L be the midpoints of AC and AB , respectively. Prove that $\angle KIL + \angle YPX = 180^\circ$.

(Proposed by Dominik Burek, Poland)

§1.27 Random Problem 3

In the diagram below, AD is perpendicular to AC and $\angle BAD = \angle DAE = 15^\circ$. If $AB + AE = BC$, find $\angle ABC$ in degrees.

§1.28 KTOM Januari 2025, Nomor 3 Esai

Diberikan persegi $ABCD$ dan titik E pada ruas CD . Misalkan F adalah titik pada ruas AE sehingga BF tegak lurus AE . Dengan cara serupa, misalkan G adalah titik pada ruas EB sehingga AG tegak lurus EB . Misalkan garis DF dan CG berpotongan di H . Buktikan bahwa

$$\angle DFC + \angle DGC = 90^\circ + \angle AHD + \angle BHC.$$

§1.29 OSN SMP 2016, Nomor 3, Hari Pertama

Pada segitiga ABC , titik P dan Q berada pada sisi BC sehingga panjang BP sama dengan CQ . $\angle BAP = \angle CAQ$ dan $\angle APB$ lancip. Apakah segitiga ABC sama kaki? Tulis alasan Anda.

§1.30 Random Problem 4

Diberikan segitiga ABC dan lingkaran luar ω dari segitiga tersebut. Ditarik sembarang garis yang sejajar BC , yang memotong sisi AB dan AC secara berurutan di titik X dan Y . Misalkan M adalah titik tengah dari busur BC pada lingkaran ω yang tidak memuat titik A . Garis MX dan MY memotong lingkaran ω kembali di titik S dan T secara berurutan. Misalkan ST dan XY berpotongan di titik P . Tunjukkan bahwa garis AP menyinggung lingkaran ω .

§1.31 A lemma from Berkeley Math Tournament 2024

The incircle of scalene triangle $\triangle ABC$ is tangent to $\overline{BC}$, $\overline{AC}$, and $\overline{AB}$ at points D , E , and F , respectively. The line EF intersects line BC at P and line AD at Q . The circumcircle of $\triangle AEF$ intersects line AP again at point $R \neq A$. If I is the incenter of $\triangle ABC$, prove that I, Q, R are collinear.

§1.32 Japan MO 2025, Problem 2

Let ABC be an acute triangle and let O be its circumcenter. Let O_1 and O_2 be the circumcenters of triangles ABO and ACO , respectively. The circumcircle of triangle AO_1O_2 intersects the side BC (excluding its endpoints) at two distinct points P and Q , such that the four points B, P, Q, C are collinear in this order. Let O_3 be the circumcenter of triangle OPQ . Prove that the three points A, O, O_3 are collinear.

§1.33 OSP SMA 2019, Essay, Problem 5

Diberikan segitiga ABC , dengan $AC > BC$, dan lingkaran luarnya yang berpusat di O . Misalkan M adalah titik pada lingkaran luar segitiga ABC sehingga CM adalah garis bagi $\angle ACB$. Misalkan Γ adalah lingkaran berdiameter CM . Garis bagi $\angle BOC$ dan garis bagi $\angle AOC$ memotong Γ berturut-turut di P dan Q . Jika K adalah titik tengah CM , buktikan bahwa P, Q, O, K terletak pada satu lingkaran.

§1.34 British MO 2019 Round 2, Problem 1

Let ABC be a triangle. Let L be the line through B perpendicular to AB . The perpendicular from A to BC meets L at the point D . The perpendicular bisector of BC meets L at the point P . Let E be the foot of the perpendicular from D to AC .

Prove that triangle BPE is isosceles.

§1.35 JBMO 2018, Problem 4

Diberikan $\triangle ABC$ dimana A', B', C' berturut-turut adalah pencerminan A, B, C terhadap BC, CA, AB . Perpotongan lingkaran luar $\triangle ABB'$ dan $\triangle ACC'$ adalah A_1 . Definisikan B_1 dan C_1 secara serupa. Buktikan bahwa AA_1, BB_1 , dan CC_1 konkuren (bertemu di satu titik).